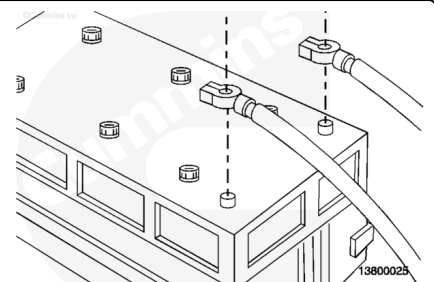


Trainings ▾

Initial Check

⚠ WARNING ⚠

Acid is extremely dangerous and can damage the machinery and can also cause serious burns. Always provide a tank of strong soda water as a neutralizing agent when servicing batteries. Wear goggles and protective clothing to avoid serious bodily injury.

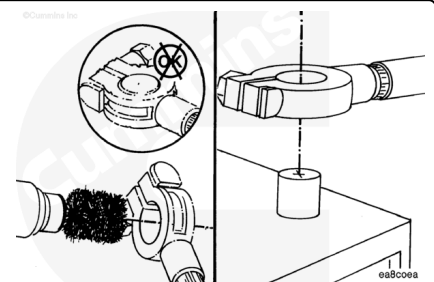


⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Label and disconnect all battery cables.

Clean corrosion and debris from battery and terminals.

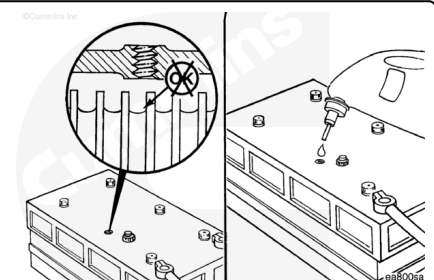


Note : Maintenance-free batteries are sealed and do not require the addition of water.

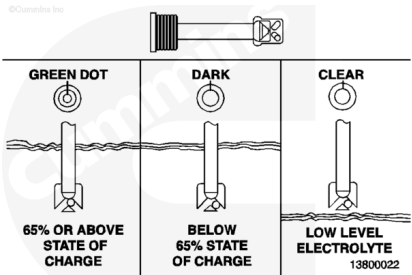
If conventional batteries are used, remove the cell caps or covers, and check the electrolyte level.

Note : If water is added to the battery it must be charged before any testing can be accomplished.

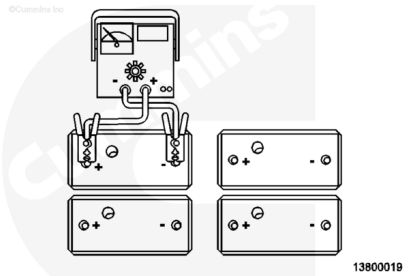
Fill each battery cell with distilled water. Refer to the battery manufacturer's specifications.



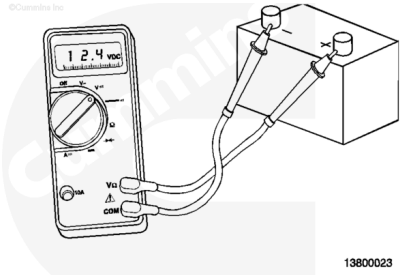
Check the “Eye” on the maintenance-free battery. Refer to the OEM specifications.



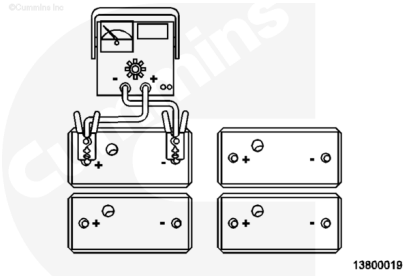
Remove the surface charge by attaching the battery to a 300-amp load for 30 seconds for heavy-duty batteries.



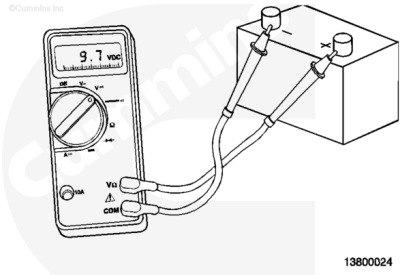
Remove the load, and wait 1 minute; if the battery voltage is greater than or equal to 12.4 VDC, continue testing. If the voltage is below 12.4 VDC, recharge or replace the battery.



Load-test the batteries at 1/2 of the cold cranking amp rating of the battery (rating at -18°C [0°F] for 15 seconds).



Check the battery voltage, and compare to the table:



Temperature and Voltage Relationship								
Temperature (F)	70	60	50	40	30	20	10	0
Temperature (C)	21	16	10	4	-1	-7	-12	-18
Minimum Voltage (VDC)	9.6	9.5	9.4	9.3	9.1	8.9	8.7	8.5

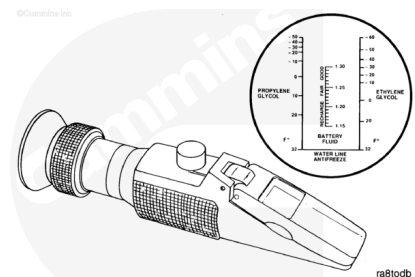
Turn the load off.

Replace the battery if it does **not** meet the above specifications.

Use the Fleetguard® Refractometer, Part Number CC-2800, to check the specific gravity of the battery electrolyte.

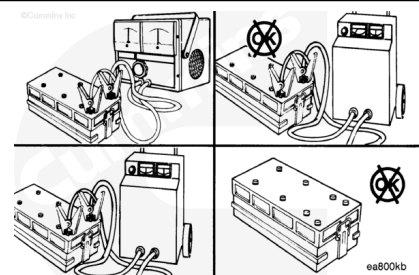
Refer to the battery fluid column in the refractometer to determine the state of charge of each battery cell.

If water has been added to a dry cell, recharge the battery to mix the added water with the existing battery electrolyte to prevent incorrect readings.



⚠ CAUTION ⚠

Do not connect battery charging cables to any electronic control system part. This can damage the electronic control system parts.



Use the systems analyzer/battery tester, Part Number 3377193, to test the output amperage of maintenance-free or conventional vent-cap batteries.

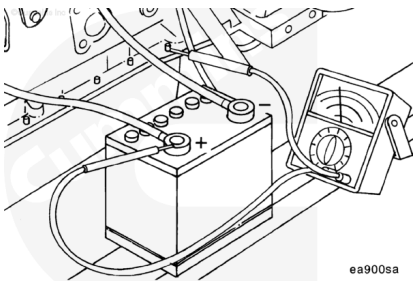
If the output amperage is low, use a battery charger to charge the battery. Refer to the manufacturer's instructions.

Replace the battery if it will **not** charge to the manufacturer's specifications or will **not** maintain a charge.

Refer to the accompanying table to determine the battery state of charge based on the specific-gravity readings.

Battery State of Charge	Specific Gravity @ 27°C [80°F]
100%	1.260 to 1.280
75%	1.230 to 1.250
50%	1.200 to 1.220

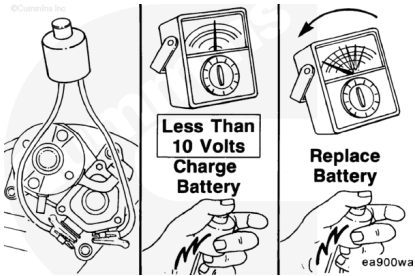
All batteries, including maintenance-free ones, can be checked by measuring the voltage between the positive (+) battery cable and the engine block ground (-). Note the voltage.



Using a remote start connection, attempt to engage the starter while observing the voltage.

If the voltage reads less than 10 VDC, charge the battery.

If the voltage drops rapidly more than 2 VDC, replace the battery.



⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Connect all battery cables.

